0580 | PAPER 4

TOPIC

10

Legacy matrices (removed from current syllabus)

Legacy practice only | Not part of the current syllabus

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SECTION 10.1

Matrices (legacy content)

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QUESTIONS

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CALCULATOR

TRACK YOUR ATTEMPTS | WRITE FULL WORKING | CHECK ACCURACY | REVIEW ERRORS

10.1 Matrices (legacy content)

#	SOURCE QUESTION	PAGES	DONE	MARK	REVIEW
01	2015 May June Paper 43 | Question 9	1	☐	___	☐
02	2016 May June Paper 43 | Question 8	1	☐	___	☐

9 $\mathbf{P} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ $\mathbf{Q} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ $\mathbf{R} = \begin{pmatrix} 0 & u \\ 1 & v \end{pmatrix}$ $\mathbf{S} = \begin{pmatrix} w & 3 \\ 8 & 2 \end{pmatrix}$

(a) Work out **PQ**.

Answer(a) $\begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}$ [2]

(b) Find $\mathbf{Q}^{-1}$.

Answer(b) $\begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}$ [2]

(c) **PR** = **RP**

Find the value of u and the value of v.

Answer(c) u = ...

v = ... [3]

(d) The determinant of **S** is 0.

Find the value of w.

Answer(d) w = ... [2]

8 $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ -1 & 5 \\ 3 & -4 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ -1 & 5 \end{pmatrix}$ $\mathbf{C} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$ $\mathbf{D} = \begin{pmatrix} 2 & 5 \end{pmatrix}$

(a) Work out each of the following if the answer is possible.
If a calculation is not possible, write "not possible" in the answer space.

(i) **BA**

[1]

(ii) 2**A**

[1]

(iii) **CD**

[2]

(iv) **DC**

[2]

(v) $\mathbf{B}^2$

[2]

(b) Find $\mathbf{B}^{-1}$, the inverse of **B**.

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

